

An Efficient Frequency and Phase Estimation Algorithm With CRB Performance for FMCW Radar Applications

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Abstract—In this paper, a chirp z -transform (CZT)-based algorithm for frequency-modulated continuous wave (FMCW) radar applications is presented. The proposed algorithm is optimized for real-time implementation in field-programmable gate arrays. To achieve a very high accuracy, the FMCW radar uses an additional phase evaluation. Therefore, a phase calculation based on the CZT algorithm is derived and compared with a correlation based algorithm. For a better classification of the algorithm, the respective Cramér–Rao bounds are calculated. The performance of the algorithm is shown by the evaluation of different radar measurements with a K-band radar. In the measurements, an accuracy of 5 μm with a mean standard deviation of 774 nm is achieved, which nearly matches the theoretically predicted mean standard deviation of 160 nm.

Index Terms—Accuracy, Cramér–Rao bounds (CRBs), frequency modulation, radar measurements, radar signal processing.

I. INTRODUCTION

TYPICALLY, frequency-modulated continuous wave (FMCW) radars for high accuracy, close-range measurements are used in industrial level measurements due to their high robustness at low costs [1]. Further, recent applications of such radars include controlling industrial machines, e.g., the piston position of hydraulic cylinders [2]. However, robust and cheap radars require a more sophisticated signal processing to achieve accuracy in the micrometer range.

An FMCW radar sends a high-frequency-modulated signal to a target and uses the received time-delayed signal to generate a low-frequency signal, the so-called beat frequency. This beat frequency is ideally a sine wave in the time domain and directly related to the position of the target. Hence, the problem of estimating a position with high accuracy reduces to the problem of estimating a spectral component (frequency and phase) in the presence of noise and distortions.

Since the hardware performance is limited, the accuracy of the FMCW radar additionally depends on the digital

analysis of its time-limited sinusoidal signal in the frequency domain. The radar signal is often distorted by interfering signals caused by multiple, unwanted reflections, and low signal-to-noise ratio (SNR) conditions. In addition, the signal should satisfy the Shannon theorem, avoid leakage with a proper windowing, and provide a sufficient spectral resolution. In other words, the spectral errors produced depend on the complex combination of many parameters such as sampling conditions, observation windows, and frequency resolution.

In contrast to these limitations, most spectral estimation methods assume a time-continuous signal, which is only distorted by white Gaussian noise. Hence, model-based parametric methods, which require a good model agreement with the real multicomponent signal and a high SNR, are unsuitable for FMCW radar measurements in demanding environments [3], [4]. Among the nonparametric frequency estimation methods, the fast Fourier transform (FFT) is the most famous method. The main drawback of this method is that if a high frequency resolution is required, a long observation window is necessary. This can either be achieved by reducing the measurement repetition rate or using zero padding. As the computational requirements are critical in radar signal processing applications, a more efficient algorithm is required. In [5], an interpolated discrete Fourier transform (DFT) algorithm is proposed. This algorithm reduces the computational requirements, but limits the choice of the observation window to a rectangular window. A different interpolated DFT algorithm, which additionally uses a Hanning window, is proposed in [6]. In general, the interpolated DFT algorithms lacks flexibility as each of these algorithms needs to be optimized separately for the chosen window function.

Another approach is to use the chirp z -transform (CZT) algorithm if only a narrow frequency band needs to be analyzed. In this case, the CZT algorithm provides good results [7] while reducing the computational requirements. In addition, the high frequency resolution of this algorithm can be further increased by the proposed modified CZT algorithm. In this paper, a CZT-based algorithm with theoretical indefinitely high-frequency resolution suitable for field-programmable gate array (FPGA) implementation and real-time applications is presented.

Manuscript received September 2, 2014; revised October 29, 2014; accepted October 30, 2014. Date of publication December 31, 2014; date of current version June 5, 2015. The Associate Editor coordinating the review process was Dr. Mark Yeary.

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Digital Object Identifier 10.1109/TIM.2014.2381354

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II. FMCW SIGNAL PROCESSING AND CZT ALGORITHM

This section presents the main principle of the FMCW radar with phase evaluation and the proposed CZT algorithm for frequency and phase calculation.

A. FMCW Radar With Phase Evaluation

Among the radar systems, the FMCW radar system with additional phase evaluation is one of the best choices, allowing absolute measurements in combination with high-distance measurement accuracy [8]. This radar system has the additional advantage that no IQ receiver is needed for the phase evaluation, keeping the overall costs of the radar system low. The FMCW radar basically works by sending a transmitter frequency f_{tx} that changes linearly with time [9] (sawtooth modulation)

$$f_{tx}(t) = f_{start} + \kappa t. \quad (1)$$

In (1), the linear frequency change with time is called chirp rate κ and is defined as $\kappa = B_{radar}/T$ with the bandwidth B_{radar} and the sweep time T . The start ramp frequency is called f_{start} . An echo signal of a reflecting object at a distance r returns after the time $\tau = 2r/c_0$ with c_0 being the speed of light. Thus, the received signal is a delayed version of the transmitted signal. If both signals are multiplied in a mixer and low-pass filtered, a low-frequency signal f_b results, which is called beat frequency

$$f_b = \frac{B_{radar}}{T} \tau. \quad (2)$$

Accordingly, the phase of the beat frequency is

$$\varphi_b = 2\pi f_{start} \tau. \quad (3)$$

Note that the frequency and the phase are directly proportional to the distance, i.e., the measurement accuracy depends on the accurate estimation of the beat frequency and the phase of the beat frequency.

It is hardly feasible to achieve accuracy in the micrometer range with the beat frequency alone as a very wide bandwidth is required [10]. Such wide bandwidth is not available in industrial, scientific, and medical radio bands [11]. A better approach is the combination of frequency and phase. This algorithm uses the phase information by dividing the entire range into segments in the order of the unambiguous range $R_{unamb} = \lambda_{start}/2$ of the phase. The position R_{coarse} is calculated via the beat frequency to identify the correct range segment. In this range segment, the phase difference $\Delta\varphi_b$ is estimated to achieve a high accuracy in the position detection (ΔR_{corr})

$$R_{coarse} = f_b \frac{c_0 T}{2 B_{radar}} \quad (4)$$

$$\Delta R_{corr} = R_{unamb} \frac{\Delta\varphi_b}{360^\circ}. \quad (5)$$

Obviously, the beat frequency and its phase, and thereby the position can be estimated by the DFT, but the CZT provides a much more efficient solution, as only a very limited part of the spectrum is necessary to estimate the position in an FMCW radar.

B. CZT Frequency Algorithm

A basic approach for estimating the position of the radar main target is the calculation of the DFT and the calculation of the maximum of the spectrum. However, the accuracy of the position detection is insufficient if the frequency grid is too coarse, i.e., the available frequency bins miss the actual maximum of the spectrum.

Starting with the DFT, another transformation with improved accuracy can be derived [12]. Through the definition of the N -point DFT

$$X_{k,DFT} = \sum_{n=0}^{N-1} x_n \cdot e^{-j2\pi \frac{nk}{N}} \quad k = 0, 1, \dots, N-1 \quad (6)$$

with time index n , discrete frequency index k , and sample length N , the normalized frequency k/N of the discrete spectrum is related to the time-continuous frequency f and the sampling frequency f_s by

$$\begin{aligned} -j2\pi \frac{nk}{N} &\stackrel{!}{=} -j2\pi f t \\ nT_s \frac{k}{N} &= f t \\ f_s \frac{k}{N} &= f \quad \text{with } f_s = \frac{1}{T_s}, \quad nT_s = t. \end{aligned} \quad (7)$$

The frequency resolution of the DFT is

$$\Delta f_{DFT} = \frac{1}{NT_s}. \quad (8)$$

If only a specific bandwidth with a certain offset of the spectrum is desired, a modified approach is necessary

$$f = f_0 + B_{CZT} \frac{k}{K} \quad \frac{k}{K} \in [0, 1]. \quad (9)$$

In (9), the frequency offset is given by f_0 . The bandwidth B_{CZT} determines the part of the spectrum, which is evaluated. The number of evaluated frequencies is determined by K . This number is independent of the available number N of samples in the time domain. Hence, the new frequency grid has a resolution of

$$\Delta f_{CZT} = \frac{B_{CZT}}{K}. \quad (10)$$

If (9) is inserted into the definition of the DFT in (6), it leads to a modified Fourier transformation

$$\begin{aligned} X_{k,CZT} &= \sum_{n=0}^{N-1} x_n \cdot e^{-j2\pi n \left(\frac{f_0 + B_{CZT} \frac{k}{K}}{f_s} \right)} \\ &= \sum_{n=0}^{N-1} x_n \cdot e^{-j2\pi \frac{f_0 n}{f_s}} e^{-j2\pi \frac{B_{CZT}}{f_s K} nk} \end{aligned} \quad (11)$$

which can be calculated after some modifications by the Bluestein FFT [12]. The Bluestein FFT corresponds to a convolution, consisting of discrete and linear filtering and is also called CZT [13]. This convolution can be implemented as fast convolution [14], which is necessary for real-time applications and the implementation of the algorithm in FPGAs. To further increase the frequency resolution and hence the accuracy of

the FMCW radar without increasing the FFT size, the CZT algorithm can be called repeatedly that performs the following.

- 1) Calculate a coarse FFT spectrum.
- 2) Estimate the maximum of the FFT spectrum.
- 3) The maximum frequency error is $\pm$ half bin width around the maximum of the FFT spectrum.
- 4) Calculate the CZT in the range of the $\pm$ half bin width of step 3) around the maximum of the FFT spectrum.
- 5) Estimate the CZT spectrum and its maximum.
- 6) Repeat steps 3) to 5), but replace the FFT spectrum with the new/updated CZT spectrum of step 5) in steps 3) and 4).
- 7) Repeat step 6) till a sufficiently high-frequency resolution is achieved.

This algorithm theoretically enables an indefinitely high-frequency resolution while keeping the computational requirements low. To compare the frequency resolution of the proposed algorithm with the frequency resolution of the DFT, a constant FFT size of $N = K$ is chosen. If the maximum frequency error of the DFT is its bin width, it follows that B_{CZT} is equal to Δf_{DFT} and (10) becomes:

$$\Delta f_{CZT, FMCW} = \frac{1}{N^2 T_s}. \quad (12)$$

Each repetition x of the algorithm further increases the frequency resolution by

$$\Delta f_{CZT, FMCW, REP} = \frac{1}{N^{(2+x)} T_s}. \quad (13)$$

Hence, even if only a short FFT is possible (e.g., using a microcontroller), a very high-frequency resolution can be achieved.

However, if the maximum of the spectrum is biased, a higher frequency resolution does not automatically allow a higher position accuracy [15]. In this case, a further improvement in position accuracy can be achieved by an additional phase evaluation algorithm.

C. CZT Phase Algorithm

To achieve highest position accuracy with the FMCW radar, the phase of the beat frequency has to be evaluated (Section II-A). An advantageous approach is to calculate the phase via the CZT, as the CZT is already used to calculate the complex frequency spectrum. For a better comparison, the CZT phase estimation is compared with the successfully implemented correlation-based phase estimation [8]. For both methods, the following discrete signal model is used:

$$x_n = A_0 \cos(2\pi k_0 n + \varphi_0) \quad \text{with} \quad k_0 = \frac{f_0}{f_s}. \quad (14)$$

In (14), x_n corresponds to a harmonic oscillation with the frequency f_0 , the sample frequency f_s , and time index n . As the phase φ_0 is estimated digitally, the following analysis is carried out using discrete-time signals.

The correlation-based phase estimation calculates the product of (14) with a complex harmonic oscillation, whose frequency $\hat{k}$ has to be previously estimated by the

CZT algorithm. After calculating the product, the signal is summed up according to

$$s_N = \sum_{n=0}^{N-1} x_n \cdot e^{-j2\pi \hat{k} n}. \quad (15)$$

The effect of errors in the frequency estimate is investigated in Section III and (30). If the frequency estimate $\hat{k}$ is identical to k_0 , the correlation can be further simplified with $k_0 = \hat{k}$ to

$$\begin{aligned} s_N &= \sum_{n=0}^{N-1} A_0 \cos(2\pi k_0 n + \varphi_0) e^{-j2\pi k_0 n} \\ &= \sum_{n=0}^{N-1} \frac{A_0}{2} e^{j\varphi_0} (1 + e^{-j2\pi 2k_0 n}). \end{aligned} \quad (16)$$

In addition to the wanted phase information, the undesired frequency $2k_0$ exists in (16), which is unavoidable for this algorithm. The equation is evaluated with the help of the partial sum over N terms of a geometric series leading to

$$s_N = \frac{A_0}{2} e^{j\varphi_0} \left(N + \underbrace{\frac{1 - e^{-j4\pi k_0 N}}{1 - e^{-j4\pi k_0}}}_{\text{systematic error term } \epsilon} \right). \quad (17)$$

The systematic error term ϵ is then simplified to

$$\begin{aligned} \epsilon &= e^{-j2\pi k_0 (N-1)} \frac{e^{j2\pi k_0 N} - e^{-j2\pi k_0 N}}{e^{j2\pi k_0} - e^{-j2\pi k_0}} \\ &= e^{-j2\pi k_0 (N-1)} \frac{\sin(2\pi k_0 N)}{\sin(2\pi k_0)}. \end{aligned} \quad (18)$$

This term becomes zero if $k_0 N$ is an integer, i.e., a multiple of the period is summed up. If the systematic error term ϵ is neglected, the phase $\hat{\varphi}$ can be determined

$$\begin{aligned} \hat{\varphi} &= \arg\{s_N\} \\ &= \arctan\left(\frac{\sin(\varphi_0)}{\cos(\varphi_0)}\right) \\ &= \varphi_0. \end{aligned} \quad (19)$$

For the general case, $\epsilon \neq 0$, the phase is distorted.

If the CZT is used for the evaluation of the phase of the beat frequency, the phase can be directly calculated out of the complex frequency spectrum (Section II-B). This algorithm first calculates the index of the maximum of (11)

$$k_{\max} = \operatorname{argmax}_k |X_{k, CZT}|. \quad (20)$$

For an easier comparison with the correlation-based phase evaluation, the index $k_{\max}$ is inserted into the initial equation (11) of the CZT algorithm. Inserting $k_{\max}$ into (11) leads to

$$X_{k_{\max}, CZT} = \sum_{n=0}^{N-1} x_n \cdot e^{-j2\pi n \left(\frac{f_0 + B_{CZT} \frac{k_{\max}}{K}}{f_s} \right)}. \quad (21)$$

The frequency found by the CZT is the frequency estimate $\hat{k}$; the exponent can therefore be simplified as

$$\hat{k} = \frac{f_0 + B_{CZT} \frac{k_{\max}}{K}}{f_s}. \quad (22)$$

If (22) is inserted into (21), the following results:

$$X_{k_{\max}, \text{CZT}} = \sum_{n=0}^{N-1} x_n \cdot e^{-j2\pi \hat{k}n} = s_N. \quad (23)$$

The phase calculated via the CZT (23) proves to be the same as the phase calculated via the correlation-based phase evaluation (15). Compared with the correlation based phase estimation, the CZT phase algorithm does not need to calculate a product with (14) and to sum up the resulting signal (15). Therefore, the computational costs are significantly reduced when the CZT algorithm is used. In summary, the CZT phase algorithm is a valid replacement of the correlation-based phase estimation and provides an efficient possibility to calculate the phase in an FMCW radar.

III. CRB AND PHASE ERROR ANALYSIS

The Cramér–Rao bound (CRB) is a lower bound for the efficiency of an estimator, which can be used to benchmark the proposed CZT algorithm. In general, the evaluation of frequency and phase of a time-limited sine wave in white Gaussian noise is an estimation problem, which is limited in its accuracy by the CRB. The lower bounds for the evaluation of frequency and phase of a sine wave are given by [15]

$$\begin{aligned} \text{var}[\hat{f}_{\text{CRB}}] &\geq \frac{12f_s^2}{(2\pi)^2 N(N^2 - 1) \cdot \text{SNR}} \\ &\approx \frac{12f_s^2}{(2\pi)^2 N^3 \cdot \text{SNR}} \end{aligned} \quad (24)$$

$$\begin{aligned} \text{var}[\hat{\phi}_{\text{CRB}}] &\geq \frac{2(2N - 1)}{N(N + 1) \cdot \text{SNR}} \\ &\approx \frac{4}{N \cdot \text{SNR}}. \end{aligned} \quad (25)$$

In (24) and (25), the sample frequency of the signal is named f_s , and the sample length is N and SNR corresponds to the signal-to-noise ratio.

Further, the CRB can be used to estimate the accuracy of an FMCW radar. In [10], the CRB is derived for the joint estimation of frequency and phase of the beat frequency of an FMCW radar. Another possibility is to derive the CRB of the phase from the frequency, which will be presented in this paper, thus implying additional conclusions regarding the influence of frequency estimation errors on the phase estimation to be drawn. As the phase estimation of the beat frequency is based on the knowledge of the exact frequency, an inaccurate frequency estimation leads to additional phase errors. To calculate the resulting phase error, (16) has to be further evaluated without the systematic error term $e^{-j2\pi 2k_0 n}$. Extending the phase with a normalized frequency error $\Delta k_e = (\Delta f_e / f_s)$ leads to

$$\tilde{s}_N = \sum_{n=0}^{N-1} \frac{A_0}{2} e^{j\phi_0 + j2\pi \Delta k_e n}. \quad (26)$$

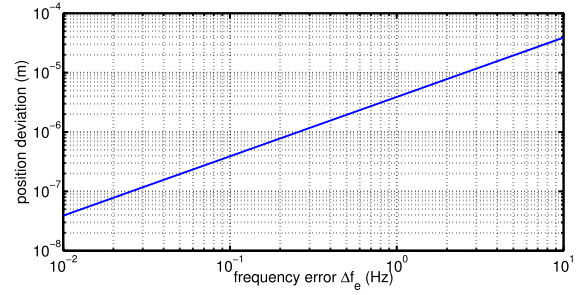


Fig. 1. Impact of frequency errors on position accuracy.

TABLE I
FMCW RADAR SENSOR SPECIFICATIONS

Operation mode	FMCW (sawtooth modulation)
Center frequency	24.3 GHz
Bandwidth	1.0 GHz (23.8 GHz to 24.8 GHz)
Ramp time	1 ms
A/D converter	NI-USB 6361 (1 MSa/s)

To perform the CZT-based phase evaluation, the geometric series has to be evaluated, resulting in

$$\begin{aligned} \tilde{s}_N &= \frac{A_0}{2} e^{j\phi_0} \frac{1 - e^{j2\pi \Delta k_e N}}{1 - e^{j2\pi \Delta k_e}} \\ &= \frac{A_0}{2} e^{j\left(\phi_0 + \frac{2\pi \Delta k_e}{2}(N-1)\right)} \frac{\sin(\pi \Delta k_e N)}{\sin(\pi \Delta k_e)}. \end{aligned} \quad (27)$$

By taking the argument

$$\begin{aligned} \tilde{\phi}_0 &= \arg\{\tilde{s}_N\} \\ &= \arg\left\{e^{j\left(\phi_0 + \frac{2\pi \Delta k_e}{2}(N-1)\right)}\right\} \\ &= \phi_0 + \underbrace{\frac{2\pi \Delta k_e}{2}(N-1)}_{\phi_e} \end{aligned} \quad (28)$$

the ideal phase with an additional error term ϕ_e results. Replacing Δk_e with $\Delta f_e / f_s$, the resulting phase error ϕ_e is

$$\phi_e = \pi \frac{\Delta f_e}{f_s} (N - 1). \quad (29)$$

The error increases linearly with the amount of samples N . This error cannot be compensated through reference measurements, as the frequency evaluation error of the CZT is random and normally distributed. The impact on the position evaluation can be calculated with help of the phase evaluation defined in (5)

$$R_e(\Delta f_e) = \frac{\Delta f_e}{4f_s} (N - 1) \cdot \lambda_{\text{start}}. \quad (30)$$

Fig. 1 shows the resulting frequency dependent position error. The frequency deviation Δf_e is varied in the range of 0.01–10 Hz. $N = 1000$ is chosen as sample length, f_s is 1 MSa/s, and f_{start} is 23.8 GHz (Table I). With a frequency error of 1 Hz (300- μm position deviation, when only the frequency is evaluated) and a sample length of N , an inherent root-mean-square position deviation of 4 μm (when the phase is

evaluated) results. In the general case, random and normally distributed errors cannot be compensated. Hence, the position error can be linked to the CRB. This variant of the CRB allows additional conclusions about the combined frequency and phase evaluation of the FMCW radar to be drawn. By combining (29) with the CRB of frequency estimation (24), the modified CRB for phase estimation results

$$\text{var}[\hat{\phi}_{m\text{CRB}}] \geq \frac{3 \cdot (N - 1)}{N(N + 1) \cdot \text{SNR}} \stackrel{N \gg 1}{\approx} \frac{3}{N \cdot \text{SNR}}. \quad (31)$$

Using (29) and (30), a position error depending on the phase can be obtained. In case of an up chirp, the following position error R_e results:

$$R_e = \frac{\lambda_{\text{start}}}{4\pi} \phi_e. \quad (32)$$

Assuming $N \gg 1$, the variance of the position error is calculated from the modified CRB as

$$\begin{aligned} \text{var}[R_e] &= \left(\frac{\lambda_{\text{start}}}{4\pi} \right)^2 \text{var}[\hat{\phi}_{m\text{CRB}}] \\ &\geq \frac{3\lambda_{\text{start}}^2}{(4\pi)^2 N \cdot \text{SNR}}. \end{aligned} \quad (33)$$

This modified CRB of the position error is linearly decreasing with $(N \cdot \text{SNR})^{-1}$.

Compared with the CRB for phase estimation of (25), the modified CRB differs by a factor of

$$\begin{aligned} \frac{\text{var}[\hat{\phi}_{m\text{CRB}}]}{\text{var}[\hat{\phi}_{\text{CRB}}]} &= \frac{3 \cdot N \cdot \text{SNR}}{4 \cdot N \cdot \text{SNR}} \\ &= \frac{3}{4} \triangleq -1.25 \text{ dB} \end{aligned} \quad (34)$$

assuming $N \gg 1$. The modified CRB is based on the assumption of an inaccurate frequency and an ideal phase not corrupted by noise. However, in a real measurement, the frequency and the phase of a sinusoid are embedded in white Gaussian noise (25). Hence, for the purpose of comparison, the CRB needs to be calculated when only the phase is distorted by noise and an ideal frequency is given. The combination of this CRB and the modified CRB will result in the generalized CRB (25). If phase estimation is performed with a single parameter, ideal frequency, and additive white Gaussian noise on the phase, the CRB is given by

$$\text{var}[\hat{\phi}_{\text{phase}}] \geq \frac{1}{N \cdot \text{SNR}} \quad (35)$$

according to [15]. The bound was simplified using $N \gg 1$. The two aspects of phase-estimation ideal phase and ideal frequency can therefore be separated. The variance of two-parameter estimation, as defined in (25), consists of

$$\text{var}[\hat{\phi}_{\text{CRB}}] = \text{var}[\hat{\phi}_{m\text{CRB}}] + \text{var}[\hat{\phi}_{\text{phase}}] \quad (36)$$

$$\frac{4}{N \cdot \text{SNR}} = \frac{3}{N \cdot \text{SNR}} + \frac{1}{N \cdot \text{SNR}}. \quad (37)$$

The impact of a frequency error on the phase accuracy is, therefore, three times as high as the impact of white Gaussian noise on the phase directly. The CRB for the position

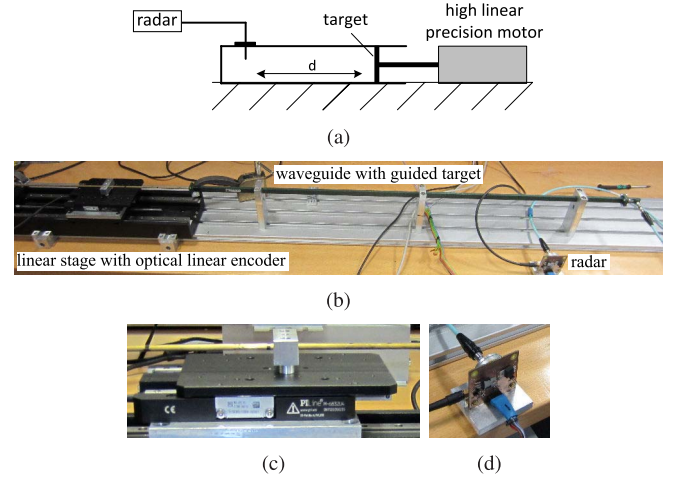


Fig. 2. Measurement setup to evaluate the performance of the CZT algorithm. (a) Schematic of the measurement setup. (b) Image of the measurement setup. (c) Linear stage with optical linear encoder. (d) Radar.

error of the two-parameter estimation in a measurement is accordingly

$$\begin{aligned} \text{var}[R_{\text{CRB}}] &= \left(\frac{\lambda_{\text{start}}}{4\pi} \right)^2 \text{var}[\hat{\phi}_{m\text{CRB}}] \\ &\geq \frac{\lambda_{\text{start}}^2}{(2\pi)^2 N \cdot \text{SNR}}. \end{aligned} \quad (38)$$

IV. EVALUATION OF THE CZT ALGORITHM BY FMCW RADAR MEASUREMENTS

Fig. 2 shows the measurement setup to evaluate the performance of the CZT algorithm. A K-band radar system with specifications according to Table I is used for the measurements. To allow the application of the CRB, some noise considerations are necessary to ensure that the measurements are mostly limited by Gaussian noise. The main source of non-Gaussian noise in a radar system is phase noise. Due to the close range measurements and the FMCW radar concept, the phase noise cancels itself out during the mixing process [16]. Furthermore, the individual radar components are designed so that they are predominantly disturbed by white noise, as other noise sources would severely affect the radar performance. Since a waveguide is used, the measurement setup is closed and has no external influences, which could cause non-Gaussian noise. However, it is also necessary to investigate the influence of deterministic errors like multiple reflections on the measurement setup.

The radar signal is coupled into a K-band waveguide [17] and an optimized short is used as radar target. This measurement setup avoids multipath scattering and is optimized for one-target scenarios. However, it is not possible to avoid all multiple reflections. Therefore, an offset of 1.40 m is chosen for the measurements. The offset ensures a stronger separation of interferences according to (2). Hence, the measurements are taken at a distance d of 1.40–1.45 m in 1 mm steps. The reference used in the

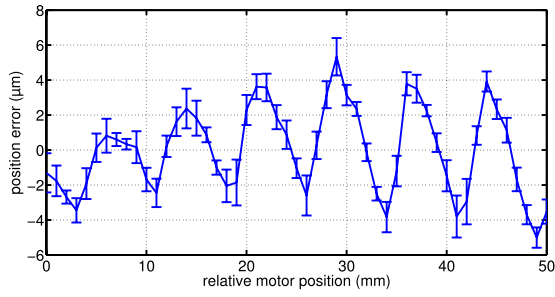


Fig. 3. Position error and standard deviation of the FMCW radar measurements.

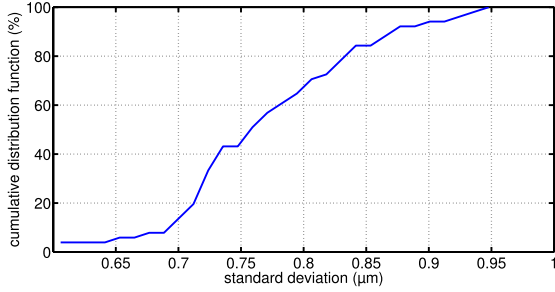


Fig. 4. Cumulative distribution function of the standard deviations. The mean standard deviation is 774 nm.

measurements is an optical linear encoder, which allows an accuracy of up to 100 nm. The target is connected to a highly linear precision motor (M-683.2U4, Physik Instrumente), which also includes the optical linear encoder.

A sampling rate of 1 MSa/s is used for all measurements. Due to the sweep time of 1 ms, there are 1000 samples in every measurement. At each measurement position, 100 measurements are taken and the complete measurement is repeated 5 times, resulting in 500 measurement values per position for statistical evaluation. Fig. 3 shows the measurement results including the respective standard deviations. An oscillation of the position error can be seen, because of the following two sources of error. As aforementioned, some multiple reflections remain in the measurement. These reflections interfere with the radar target and lead to a position dependent error. Similarly, the image frequency affects the position error for very high-position accuracies.

For comparison with the CRB, the cumulative distribution function of the standard deviations is calculated and shown in Fig. 4. The mean standard deviation of all position measurements is 774 nm.

To compare the measurement results with the theoretical CRB prediction, it is also necessary to estimate the SNR of the FMCW radar. Hence, the spectrum is calculated via an FFT and interpolated by zero padding. To reduce side lobes, a Blackman window is applied. The resulting spectrum is shown in Fig. 5. As the used FMCW radar is a monostatic radar, it is not possible to make a direct connection between the input and the output of the radar for measuring the SNR of the radar. It is also not possible to measure the noise floor by terminating the waveguide with a matched load. A matched termination would neglect the noise of the radar itself, which also gets reflected into the radar in a real

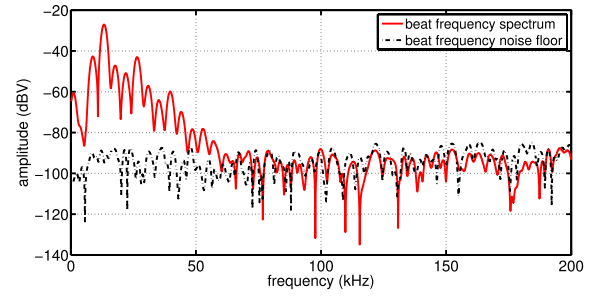


Fig. 5. Amplitudes of the beat frequency for estimating the SNR of the FMCW radar.

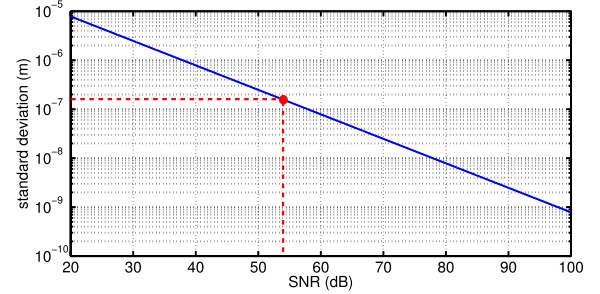


Fig. 6. Cramér–Rao bound of the used FMCW radar.

measurement. However, the noise floor can be calculated if the same measurement is taken twice and both measurements are subtracted from each other. As the white noise possesses a random distribution, only nonrandom signals get removed from the spectrum by this method and the noise remains. The resulting noise floor is also shown in Fig. 5. The noise power is calculated by the variance of the amplitude of the beat frequency in the range of 8–100 kHz. This frequency range corresponds to the 3-dB beat frequency bandpass filter bandwidth in the FMCW radar. An approximated SNR_{FFT} of 72.5 dB is calculated. For comparison with the CRB in the time domain, it is necessary to subtract the processing gain of the FFT (27 dB) and to add the loss of the Blackman window (8.5 dB, [18]), which results in an SNR_{time} of 54 dB. Fig. 6 shows the CRB, which is calculated via (38). The sample length is set to $N = 1000$. The wavelength λ_{start} is set to the corresponding start frequency f_{start} of 23.8 GHz. For the calculated SNR_{time} of 54 dB, a mean standard deviation of about 160 nm can be expected. The calculated CRB is the best case estimation and only considers white noise. Considering the additional sources for errors in the measurements due to hardware imperfections (multiple reflections and imperfect reflection at the target nonperfect interconnects), the measured mean standard deviation of 774 nm seems to be appropriate when compared with the theoretical value.

V. CONCLUSION

An optimized CZT algorithm for FMCW radar measurements is presented. It is shown that the algorithm can be applied for phase estimation as well, because it provides the same results, as obtained with the computationally intensive correlation-based method. To benchmark the proposed

algorithm, the respective CRB is derived. In addition, the relation between frequency estimation errors and phase estimation errors is shown. It can be concluded that the influence of frequency errors on the phase accuracy of the FMCW radar is three times as high as the influence of white noise on the phase directly. Hence, an algorithm with advantageous frequency accuracy, i.e., the CZT algorithm, is necessary to achieve high accuracy with an FMCW radar. Furthermore, a method to measure the SNR of a monostatic FMCW radar is presented and it is shown that it is possible to predict the standard deviation of the accuracy of FMCW radar measurements with the help of the CRB. The achieved accuracy of 5 μm with a mean standard deviation of 774 nm proves the performance of the proposed algorithm.

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